Project-LowLevelDesign

On

Multimodal Manufacturing Creator

Course Name: Generative AI

Institution Name:Medicaps University – Datagami Skill Based Course

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1. Introduction

The **Multimodal Manufacturing Creator** is a Generative AI-based system that transforms textual manufacturing ideas into **comprehensive outputs**, including:

- Detailed product descriptions
- Functional explanations
- High-quality visual prototypes

This system integrates:

- **Large Language Models (LLMs)** → for text generation
- **Image Generation Models (via API)** → for visualization
- **Vector Database** → for semantic retrieval and context enhancement

The goal is to enable users to quickly **visualize, understand, and present manufacturing concepts** without needing complex design tools like CAD software.

1.1 Scope of the Document

This document provides a detailed low-level design of the Multimodal Manufacturing Creator, including:

- Multimodal AI workflow (text + image generation)
- Integration of LLM APIs and Image APIs
- Prompt processing and enhancement
- System architecture and execution flow
- Backend services and database design

The scope also includes implementation details such as:

- Prompt engineering pipeline
- Vector database usage for context retrieval
- API structure for generation modules
- Storage of generated outputs

1.2 Intended Audience

This document is intended for:

- Software Developers
- AI/ML Engineers
- System Designers
- Project Evaluators

- Students working on Generative AI projects

1.3 System Overview

The system allows users to:

- Input manufacturing ideas (e.g., *"Eco-friendly smart water bottle with temperature control"*)
- Automatically generate:
  - Detailed product descriptions
  - Design specifications
  - Realistic product images

Key Features

- Multimodal generation (text + image)
- Prompt enhancement using LLM
- Semantic search using Vector DB
- Real-time visualization
- Download/export outputs

2. System Design

2.1 Application Design

Architecture Type

Client-Server Architecture with AI Integration

System Layers

1. Frontend Layer

- Web interface (React / HTML / JS)
- Input field for prompts
- Output display (text + images)
- User dashboard

2. Backend Layer (Python)

- Handles business logic
- API orchestration
- Prompt processing

3. AI Layer

- LLM API (text generation)
- Image Generation API
- Prompt Engineering Module

4. Data Layer

- Vector Database (semantic retrieval)
- Storage database (user data, outputs)

2.2 Process Flow

Step-by-Step Workflow

1. User logs into system
2. User enters manufacturing concept
3. Backend receives input
4. Query sent to Vector DB:
    - Retrieve related manufacturing knowledge
5. Prompt is enhanced using LLM
6. Enhanced prompt is used for:
    - Text generation (LLM)
    - Image generation (API)
7. Results are returned to backend
8. System displays:
    - Description
    - Generated image
9. Output stored in database

2.3 Information Flow

Input

- User prompt
- Optional parameters (style, material, industry)

Processing

- Semantic search via Vector DB
- Prompt enhancement using LLM
- Parallel generation:
    - Text generation
    - Image generation

Output

- Product description
- Functional explanation
- Generated prototype image
- Suggestions/improvements

2.4 Components Design

1. Frontend Components

a. User Interface (UI)

- Prompt input
- Output display
- Navigation dashboard

b. Visualization Module

- Displays generated images
- Zoom / download options

c. Input Control Module

- Accepts structured/unstructured prompts
- Allows parameter selection

d. Authentication Module

- Login / Register
- User session management

2. Backend Components (Python)

a. API Layer

- Receives frontend requests
- Routes to services

b. Prompt Engineering Module

- Cleans and structures input
- Enhances prompt using LLM
- Combines Vector DB context

c. Vector Search Module

- Uses embeddings
- Retrieves relevant manufacturing knowledge
- Improves prompt quality

d. LLM Module

- Generates:
  - Product description
  - Design explanation
  - Use-case details

e. Image Generation Module

- Sends prompt to image API
- Generates prototype visuals

f. Multimodal Orchestrator

- Synchronizes text and image outputs
- Ensures consistency

g. Storage Module

- Stores:
  - Prompts
  - Generated text

- Image URLs

h. User Management Service

- Handles authentication
- Stores user history

2.5 Key Design Considerations

1. Accuracy

- Use Vector DB + LLM for better context
- Reduce hallucinations

2. Scalability

- Modular microservice design
- Easy integration with new AI models

3. Performance

- Parallel API calls (text + image)
- Caching frequently used prompts

4. Security

- Secure API keys
- User authentication

5. User Experience

- Fast response time
- Clean UI
- High-quality visuals

6. Cost Optimization

- Limit API calls
- Efficient prompt engineering

2.6 API Catalogue

1. Prompt Enhancement API

- Endpoint: /enhancePrompt
- Method: POST
- Input: User prompt
- Output: Enhanced prompt

2. Text Generation API

- Endpoint: /generateText
- Method: POST
- Input: Enhanced prompt
- Output: Product description

3. Image Generation API

- Endpoint: /generateImage
- Method: POST
- Input: Enhanced prompt
- Output: Image URL

4. Multimodal API

- Endpoint: /generateAll
- Method: POST
- Input: User prompt
- Output:
    - Description
    - Image

5. Vector Search API

- Endpoint: /searchContext
- Method: POST
- Input: Prompt
- Output: Related data

6. User APIs

- /login
- /register
- /history

3. References

- OpenAI / LLM API Documentation
- Stable Diffusion / Image Generation APIs
- Vector Database Documentation (FAISS / Pinecone)
- Generative AI Course Materials